



DEPARTMENT OF THE ARMY
US ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND
GROUND VEHICLE SYSTEMS CENTER
6501 E. 11 MILE ROAD
WARREN, MI 48397-5000

**COOPERATIVE RESEARCH AND DEVELOPMENT AGREEMENT
BETWEEN
TO THE STARS ACADEMY OF ARTS AND SCIENCE, INC.
AND
THE U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND
GROUND VEHICLE SYSTEMS CENTER**

**CRADA TITLE: NOVEL & EMERGING TECHNOLOGY EXPLOITATION (NETE)
CRADA NUMBER: 19-15**

CRADA ADMINISTRATORS:

GOVERNMENT

PRINCIPAL INVESTIGATOR:

**Ben Soave
Ground Vehicle Survivability and Protection (GVSP)
Mailing address:**

**CCDC Ground Vehicle Systems Center
6501 E. 11 Mile Road, MS 263b
Warren, MI 48397-5000**

**Dr. Joseph Cannon
Ground Vehicle Survivability and Protection (GVSP)
Mailing address:**

**CCDC Ground Vehicle Systems Center
6501 E. 11 Mile Road, MS 263b
Warren, MI 48397-5000**

OFFICE OF RESEARCH AND TECHNOLOGY APPLICATIONS ("ORTA"):

**Andrea Simon
Office of Research & Technology Application
Mailing address:**

**CCDC Ground Vehicle Systems Center
6501 E. 11 Mile Road, MS 221
Warren, MI 48397-5000**

LEGAL COUNSEL:

**Christina Degnore
Chief, Intellectual Property Law
Division Mailing address:**

**US Army TACOM LCMC
6501 East Eleven Mile Road, AMSTA-LP / MS 281
Warren, Michigan 48397-5000**

COLLABORATOR

PRINCIPAL INVESTIGATOR:

**Luis Elizondo
Director of Global Security and Special Programs
Mailing address:**

**315 South Coast Hwy 101
Ste. U-38
Encinitas, CA 92024**

PREFERRED CONTACT:

**Luis Elizondo
Director of Global Security and Special Programs
Mailing address:**

**315 South Coast Hwy 101
Ste. U-38
Encinitas, CA 92024**

LEGAL COUNSEL:

**Heidy Vaquerano, Esq.
Wolf, Rifkin, Shapiro, Schulman & Rabkin, LLP.
Mailing address:**

**11400 W. Olympic Boulevard, 9th Floor
Los Angeles, CA 90064**

SECTION II – JOINT WORK PLAN

NOVEL & EMERGING TECHNOLOGY EXPLOITATION (NETE)

ARTICLE A - PROJECT DESCRIPTION

To the Stars Academy of Arts and Science is a company with materiel and technology innovations that offer capability advancements for Army ground vehicles. These technology innovations have been acquired, designed, and produced by the Collaborator, leveraging advancements in metamaterials and quantum physics to push performance gains. The Collaborator will share their innovations with the Government, who will use its laboratories and resources to characterize the technologies and determine if they have appropriate applications on ground vehicles.

ARTICLE B - OBJECTIVES

B.1. CRADA Objective.

The Collaborator has access to advanced materiel solutions. The Government would like to assess, test, and characterize the products from the Collaborator at Government facilities to compare the capabilities of these advancements to known commodities, understand what facilities would be required to reproduce the advancements, and determine their applications for ground vehicle platforms.

B.2. Benefit to Government.

If the Government can verify materiel solutions claims by the Collaborator, then significant advancements can be made in the capabilities of Army ground vehicle platforms in terms of security, force protection and weight reduction.

B.3. Benefit to Collaborator.

The collaboration described within this document enables the Collaborator to understand the performance of their materiel innovations in Army ground vehicle environments and benchmark that capability against existing technology capabilities. This CRADA will benefit the Collaborator by providing guidance, recommendations, additional data and results collected from Government testing.

ARTICLE C - PARTIES AND OTHER PARTICIPANTS

C.1. Relationship of Parties.

The Collaborator contacted Government about the innovations within their hold and proposed joint research. The Collaborator is a public benefit company with limited resources. Collaborator leaders are prior Government employees who understand the value that materiel solutions can provide for the Warfighter. Government personnel recognize the value of the technologies described and can provide Government facilities to assess, test, and characterize the technologies for better understanding the performance described and compare them against accepted standards.

C.2. Other Participants

The other contributors for this effort are:

- 1) The Office of the Secretary of Defense (OSD) can share historical reports of findings and origin of materiel solutions in the possession of the Collaborator.

ARTICLE D - TECHNICAL TASKS AND ASSOCIATED RESOURCES

D.1. Government Activity.

Government shall:

1. Perform assessments, testing, and characterization of Collaborator-provided technologies. The Government is interested in a variety of the Collaborator's technologies, such as, but not limited to inertial mass reduction, mechanical/structural metamaterials, electromagnetic metamaterial wave guides, quantum physics, quantum communications, and beamed energy propulsion.
2. Provide Subject Matter Expertise technical support, guidance, evaluation, and temporary duty, as required and at the discretion of the Government.
3. Share all unclassified test results pertaining to Collaborator technologies in compliance with applicable SCGs.
4. Author report(s) to document the testing conducted and the performance achieved during those tests.
 - a) The Government will share the reports with the Collaborator at the conclusion of this effort.
5. Sponsor all appropriate security clearances necessary for work under this effort.
 - a) Provide necessary clearance to appropriate classified material(s) through a DD254.
6. Return Collaborator-provided materials at the conclusion of testing and evaluation.
7. The total value of anticipated USG resources, including personnel, services provided, Intellectual Property, facilities, equipment, and reporting through the course of this effort is estimated at \$750k.
8. Secure storage of technology and artifacts of sensitive nature.

D.2. Collaborator Activity.

Collaborator shall:

1. Provide SMEs in the designated areas of study, including physics, material science, and field collections.
2. Share knowledge, exploratory research, pertinent internal reports, and results (reports) of prior testing with respect to material, material science, and advanced technologies with USG personnel.
3. Provide the agreed upon materiel technologies for USG testing and evaluation.
 - a) Metamaterial: Samples of mechanical and EM sensitive metamaterial collected, obtained, or developed as part of its Field Operations
 - b) Material Analysis: Written data, information, and analysis related to tested materials
 - c) Quantum Communication:
 - 1) Related research to-date on theories, studies, mathematical formulas, and prototypes

- 2) Future developments, prototypes, and hardware associated with the specific quantum product
- d) Beamed Energy Propulsion
 - 1) Launch vehicles, vehicle prototypes, and systems obtained or possessed
 - 2) Data associated with testing, developing, and improving persistence and stability of launch vehicles or launch systems
 - 3) Proposed application of these systems
- e) Active Camouflage and Directed Photon Projection
 - 1) Projection systems
 - 2) Technology associated with these systems
 - 3) Material projection surfaces
 - 4) Proposed applications of these systems
4. Provide secure shipping of Collaborator technologies to the Government.
5. The total value of the Collaborator's participation (personnel, services, Intellectual Property, facilities, equipment or other resources) is estimated at \$1M, but will be determined by business needs and funding availability.

D.3. Joint Activity.

Both Parties shall jointly:

1. Share current knowledge on existing material and available data (if any).
2. Use available resources to design and prepare samples testing/evaluation where appropriate
3. Develop test and evaluation plans for material or equipment under evaluation.
4. Consolidate testing/evaluation results and include in a jointly developed report that will be provided to both the Collaborator and Government. This reporting shall summarize testing conducted, characterizations of the technology performance, and suggested next steps or opportunities for the technologies.

D.4. Property and Equipment.

Unless specified in paragraphs D.1 and D.2, no property nor equipment will be furnished by either party under this CRADA.

ARTICLE E - INTELLECTUAL PROPERTY

E.1. Background Technology.

E.1.1. Collaborator's Background Technology.

Collaborator's prior research, discoveries, and proprietary designs for the materiel solutions provided to Government are exclusive property of the Collaborator. That information shall remain owned by the proprietor with the rights by the U.S. Government for the testing described within this document.

E.1.2. Government's Background Technology.

None.

E.2. Marking of Background Technology.

All Background Technology will be identified as such with a marking. For example:

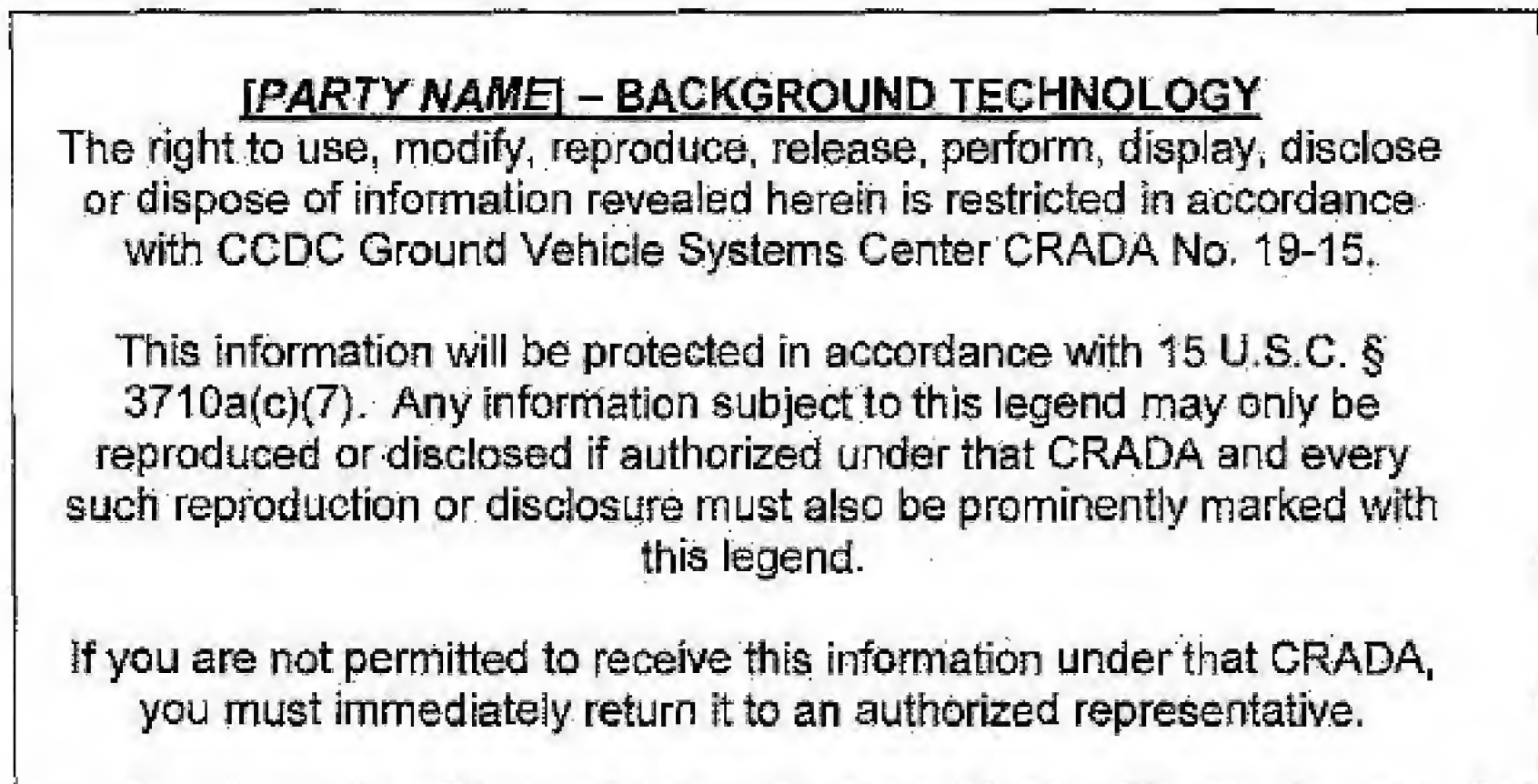


Fig 1. Marking of Background Technology

E.3. Marking of Proprietary Information.

All Proprietary Information will be identified as such with a marking. For example:

[PARTY NAME] – PROPRIETARY INFORMATION

The right to use, modify, reproduce, release, perform, display, disclose or dispose of information revealed herein is restricted in accordance with CCDC Ground Vehicle Systems Center CRADA No. 19-15. This information will be protected in accordance with 15 U.S.C. § 3710a(c)(7). Any information subject to this legend may only be reproduced or disclosed if authorized under that CRADA and every such reproduction or disclosure must also be prominently marked with this legend.

If you are not permitted to receive this information under that CRADA, you must immediately return it to an authorized representative.

Fig 2. Marking of Proprietary Information

ARTICLE F - DELIVERABLES

F.1. Reports.

At the conclusion of each fiscal year, the Government shall complete reporting of testing conducted, inclusive of characterization of the technologies, findings, observations, and recommendations. The Collaborator shall contribute to the reports with their scientist analysis. Any CRADA Inventions Made and other Intellectual Property developed or Created, including Works, and their respective owners and licensees, shall be identified and agreed to by all parties involved. Government and Collaborator have the opportunity to request additional detail from the Report preparer, if it is deemed necessary to better understand the technical progress made, identify problems encountered, and establish goals and objectives requiring further effort. The reports shall be authored in Microsoft Word in a Government format. Final drafts shall be converted to PDF. All reports delivered under this CRADA shall be shared with all parties involved, subject to appropriate class guides.

F.2. Other Deliverables.

No additional deliverables are anticipated within the scope of this CRADA.

ARTICLE G - MILESTONES

Activity	Start	End	Reference to Joint Work Plan	Notes
Technical Work	10/1/2019	30 Sep 2023	D.1.1, D.1.6	
Milestone/Review FY20	9/1/2020	9/30/2020	D.1.1	
IP Review FY20	9/1/2020	9/30/2020	D.1.6	
Milestone/Review FY21	9/1/2021	9/30/2021	D.1.1	
IP Review FY21	9/1/2021	9/30/2021	D.1.6	
Milestone/Review FY22	9/1/2022	9/30/2022	D.1.1	
IP Review FY22	9/1/2022	9/30/2022	D.1.6	
Milestone/Review FY23	9/4/2023	9/29/2023	D.1.1	
IP Review FY23	9/4/2023	9/29/2023	D.1.6	
CRADA Closeout	9/4/2023	9/29/2023		
Other Activities				
- Monthly communications				
- Tests				
- Status Reports				